

Development of an IoT-Based Smart Sub-Metering System for Room-Level Electricity Monitoring, Carbon Emission Estimation, and Fair Electricity Cost Allocation

1 Introduction

Shared residential spaces often rely on a common electricity meter and divide costs equally among occupants, even when their actual electricity consumption differs. Occupants may use different appliances, follow different daily routines, and have different economic circumstances. As a result, equal bill sharing may not accurately reflect individual electricity use or the financial burden of electricity consumption.

IoT-based smart sub-metering provides a practical approach for measuring electricity consumption at the individual room level. Previous studies have shown substantial differences in dormitory electricity consumption due to occupant behavior, room characteristics, and occupancy patterns [1,3,5]. IoT-based systems have also demonstrated the feasibility of continuous, high-resolution electricity monitoring and real-time visualization [2,6,8]. However, existing studies have generally focused on electricity monitoring, energy management, or system development separately, with less attention to combining room-level consumption measurement with carbon-emission estimation and electricity-cost allocation.

This study develops an IoT-based smart sub-metering system integrated with a web-based monitoring platform to address this gap. The system measures room-level electricity consumption and uses these measurements to estimate carbon emissions and electricity costs. In addition, economic position is considered as an additional factor when evaluating electricity-cost fairness. The study involves two working occupants with different economic positions: a journalist residing in Room K and a medicine company representative residing in Room F with a comparatively lower economic position.

The study therefore examines differences in electricity consumption, carbon emissions, and electricity cost burden between the two occupants and considers how these differences relate to their economic circumstances. The developed web-based platform provides a practical interface for monitoring, visualizing, and analyzing the collected electricity data.

1.1 Novelty of the Study

The main contributions of this study are:

- Development of an IoT-based room-level electricity monitoring system integrated with a web-based platform.
- Estimation of individual carbon emissions from measured electricity consumption.
- Comparison of conventional equal-bill and consumption-based electricity cost allocation.
- Inclusion of economic position as an additional factor when evaluating electricity-cost fairness.

1.2 Background

Electricity consumption in shared residential environments varies between occupants because of differences in appliance use, daily routines, and occupancy patterns. Previous dormitory studies have found substantial differences in room-level electricity consumption and identified factors such as appliance use, room location, and academic schedules as important influences [1,3,5]. These findings support the need for more detailed monitoring than a single shared electricity meter can provide.

IoT-based smart metering can address this problem by providing continuous electricity measurements at the room or equipment level. Studies have demonstrated the feasibility of collecting high-resolution electricity data and using IoT systems for real-time monitoring and energy management [2,6,8]. Smart monitoring platforms have also been shown to improve energy awareness and support energy-management decisions [4,7].

However, conventional shared billing generally records only the total electricity consumption of a residence. It does not show how much electricity each occupant used, the carbon emissions associated with that use, or whether each occupant's share of the bill reflects their actual consumption. Room-level monitoring can address these issues by linking individual electricity use with cost and environmental impact.

Fairness may also depend on economic circumstances. Two occupants paying the same amount may experience different financial burdens if their economic positions differ. Therefore, considering economic position alongside measured electricity consumption provides an additional perspective when evaluating electricity-cost fairness.

This study applies IoT-based sub-metering to two working occupants with different economic positions and uses a web-based platform to monitor their electricity use. The system measures consumption, estimates carbon emissions, and compares equal-bill with consumption-based cost allocation. In this way, the study extends previous work on electricity monitoring by combining consumption, environmental impact, and cost allocation at the room level.

1.3 Research Objectives

The aim of this study is to develop and evaluate an IoT-based smart sub-metering system for individual electricity monitoring, carbon emission estimation, and fair electricity cost allocation while considering occupants' economic positions.

The specific objectives are:

1. To develop an IoT-based smart sub-metering system for room-level electricity monitoring.
2. To measure and analyze individual electricity consumption using the developed system.
3. To estimate individual carbon emissions from measured electricity consumption.
4. To compare equal-bill and consumption-based electricity cost allocation.

1.4 Research Questions

1.4.1 Main Research Question

How can an IoT-based smart sub-metering system support individual electricity monitoring, carbon emission estimation, and fair electricity cost allocation while considering occupants' economic positions?

1.4.2 Sub-Research Questions

RQ1: How does electricity consumption differ between the two occupants based on room-level measurements?

RQ2: What are the corresponding carbon emissions from each occupant's electricity consumption?

RQ3: How does consumption-based electricity allocation differ from equal-bill allocation?

RQ4: How do differences in economic position relate to electricity cost burden?

RQ5: How can the developed web-based system support monitoring and visualization of electricity consumption, carbon emissions, and costs?

1.5 Limitations of the Study

This study is limited to two occupants and a 15-day monitoring period, so the findings cannot be generalized to other shared residential environments or economic groups. The economic-position comparison is limited to these occupants and indicates an association rather than a causal relationship with electricity consumption. The study also considers only electricity-related carbon emissions and does not include emissions from other household activities or sources.

2 Methodology

This study used a quantitative, IoT-based observational approach to measure room-level electricity consumption, carbon emissions, and electricity costs. Quantitative observation was appropriate because the study relied on numerical measurements collected systematically over time. Manual meter readings and questionnaire-based approaches were considered, but manual readings provide fewer observations and require repeated human intervention, while questionnaires depend on participants' ability to accurately report their electricity use. Automated IoT sub-metering was therefore selected because it provided direct, repeated measurements and was better suited to comparing electricity consumption between individual rooms.

2.1 Data Collection Methods

Electricity data were collected from Room F and Room K over a 15-day period using two Tuya-compatible smart meters. The meters communicated with the monitoring system through the local Wi-Fi network using the TinyTuya Python API, without relying on Tuya Cloud.

Measurements were collected every 10 seconds. Each successful reading recorded voltage (V), current (A), instantaneous power (W), cumulative energy consumption (kWh), device status, switch status, fault status, and timestamp. The readings were stored in a local SQLite database for analysis.

Automated measurement was preferred because it provided actual high-frequency electricity data rather than relying on estimated or self-reported consumption. This approach is consistent with previous IoT energy-monitoring studies that used automated sensing to provide continuous electricity measurements [2,6,8]. However, the system was locally hosted and depended on a computer and network connection remaining available. Power outages or network interruptions could therefore interrupt data collection and require the system to be restarted.

The developed web-based system, built using Flask, Pandas, NumPy, SQLite, and TinyTuya, was used to monitor the devices and visualize the collected data. It provided live monitoring, historical consumption trends, room comparisons, electricity-cost analysis, and carbon-emission estimates.

2.2 Participants / Sampling

The study used purposive sampling to select two suitable rooms, Room F and Room K, occupied by two working individuals with different economic positions. The rooms were selected based on their suitability for installing and operating the smart-metering system.

Purposive sampling was appropriate because the study aimed to obtain detailed measurements from specific rooms rather than produce a statistically representative sample of all residential occupants. The resulting study was therefore treated as a small 15-day case study. This allowed detailed comparison between the two rooms but limited the generalizability of the findings to other dormitories and economic groups.

3 Task 06. Data Analysis Plan

3.1 Data Preparation Plan

The study will collect quantitative electricity data from IoT-based smart submeters installed in the selected rooms. The collected data will include electricity consumption (kWh), voltage (V), current (A), active power (W), and the date and time of each measurement. Additional information, including anonymous room identification, number of occupants, and the total electricity bill, will also be collected.

Before analysis, the collected dataset will be checked for missing, duplicated, or invalid records. Timestamps will be standardized, and the measurements will be organized by room and time period. Any measurements

affected by Internet interruptions or load-shedding will be identified so that the reliability of local storage and later synchronization can also be examined.

After cleaning, the data will be stored and processed in a structured format using SQLite and Python. Measurements will be grouped by room and by appropriate time periods, such as daily and hourly intervals, to support comparison between rooms. This preparation is consistent with the importance of checking and organizing quantitative data before analysis discussed by Denscombe (2010, pp. 235–240).

3.2 Initial Exploration Plan

The first stage of analysis will use descriptive statistics and graphical exploration to understand the collected electricity data. For each room, the analysis will examine:

- Total electricity consumption (kWh)
- Average electricity consumption
- Mean and median power
- Standard deviation
- Minimum and maximum power
- Voltage and current patterns
- Daily and hourly electricity-use patterns

The data will also be examined for unusual values and noticeable differences between the monitored rooms. Daily and hourly summaries will be used to identify periods of higher or lower electricity demand.

Graphs will be used during this stage to make differences and patterns easier to identify. This approach follows the use of descriptive and exploratory techniques for understanding quantitative datasets discussed by Denscombe (2010, pp. 235–240).

3.3 Analysis Plan

The analysis will be conducted quantitatively and will be organized around the research questions.

3.3.1 Electricity Consumption

Room-level electricity consumption will be compared using descriptive measures such as total, mean, median, minimum, maximum, and standard deviation. Measurements will be grouped by room and time period to identify differences in overall consumption and power-demand patterns.

The percentage difference in consumption between rooms will also be calculated. Daily and hourly electricity-use patterns will be examined to determine whether consumption differs during particular periods.

3.3.2 Carbon Emission Estimation

Carbon emissions will be estimated from the measured electricity consumption using the selected Bangladesh grid emission factor. The planned calculation is:

$$C_{\text{emission}} = E_{\text{electricity}} \times EF_{\text{grid}} \quad (1)$$

where C_{emission} represents estimated carbon emissions, $E_{\text{electricity}}$ represents measured electricity consumption in kWh, and EF_{grid} represents the applicable grid emission factor.

The estimated carbon emissions for each room will then be compared to determine how differences in electricity consumption correspond to differences in estimated environmental impact.

3.3.3 Electricity Cost Allocation

Two billing methods will be compared.

For equal-bill allocation:

$$\text{Cost per Room} = \frac{\text{Total Electricity Bill}}{\text{Number of Rooms}} \quad (2)$$

For consumption-based allocation:

$$\text{Room Cost} = \frac{\text{Room Electricity Consumption}}{\text{Total Electricity Consumption}} \times \text{Total Electricity Bill} \quad (3)$$

The analysis will compare the amount assigned to each room under the two methods. The absolute and percentage differences between the two approaches will be calculated to show whether consumption-based allocation produces a different distribution of the electricity cost.

3.3.4 System Reliability

The collected data will also be examined to assess whether the monitoring system can maintain useful data collection during Internet interruptions and load-shedding. The analysis will consider the availability of measurements, locally stored records, recovered data, and synchronization after connectivity is restored.

The results will be used to assess whether the system can support reliable room-level monitoring under the infrastructure conditions considered in the study.

3.4 Presentation Plan

The findings will be presented using tables, graphs, and numerical summaries.

The planned presentation will include:

- A summary table of the collected electricity variables
- Room-wise electricity consumption comparisons
- Daily and hourly power-consumption graphs
- Carbon-emission comparisons between rooms
- Equal-bill versus consumption-based billing comparisons

Tables will be used when exact numerical values are important, while graphs will be used to show trends and differences between rooms. Results will be reported using appropriate units, including kWh for energy, W for power, V for voltage, A for current, kg CO₂ for estimated emissions, and BDT for electricity cost.

The presentation will focus on connecting each analysis outcome to the corresponding research question rather than presenting the collected data without interpretation. This follows Denscombe's discussion of organizing and presenting quantitative data in a way that supports meaningful interpretation (2010, pp. 241–259).

References

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